Diff of Exponential & logarithmic function

(指數及對數函數微分)

Rule No. 1 :

$$\frac{d}{dx}(e^x) = e^x$$



e.g. ***1.*** $\frac{d}{dx}(e^2) =$

2. $\frac{d}{dx}(e^x) =$

3. $\frac{d}{dx}(2e^x) =$

Rule No. 2 : **Given *k* is a constant (常數),**

$$\frac{d}{dx}(e^{kx}) = ke^{kx}$$

e.g. ***1.*** $\frac{d}{dx}(e^{2x}) = 2e^{2x}$



2. $\frac{d}{dx}(e^{-3x}) =$

3. $\frac{d}{dx}(\frac{1}{e^x}) =$

4. $\frac{d}{dx}(e^x)^2 =$

5. $\frac{d}{dx}(e^x + \frac{1}{e^x}) =$

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Rules of Differentiation (Exponential Function)

Rule No. 3 : Chain Rule in differentiate Exponential function

$$\frac{d}{dx}(e^{f(x)}) = \frac{d\,(e^{f(x)})}{d\,f(x)} \cdot \frac{d\,f(x)}{dx} = f'(x)\cdot e^{f(x)}$$

e.g. ***1.*** $\frac{d}{dx}(e^{x^2}) = \left(\frac{d}{dx}x^2\right)\cdot e^{x^2} = 2xe^{x^2}$

2. $\frac{d}{dx}(e^{2x-1}) = \left[\frac{d}{dx}(2x-1)\right]\cdot e^{2x-1} = 2e^{2x-1}$



3. $\frac{d}{dx}(e^{3x^2+5x}) =$

4. $\frac{d}{dx}(e^{\frac{1}{x}}) =$

5. $\frac{d}{dx}(e^{\sqrt{x}}) =$

6. $\frac{d}{dx}(e^{x+\frac{1}{x}}) =$

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Rules of Differentiation (logarithmic function)

Rule No. 4 : **For $x > 0$,**

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$



e.g. ***1.*** $\frac{d}{dx}(\ln 2) =$

2. $\frac{d}{dx}(\ln x) =$

3. $\frac{d}{dx}(2\ln x) =$

Rule No. 5 : **Given k is a positive constant and $x > 0$,**

$$\frac{d}{dx}(\ln kx) = \frac{d}{dx}(\ln k + \ln x) = \frac{1}{x}$$

e.g. ***1.*** $\frac{d}{dx}(\ln 2x) = \frac{1}{x}$



2. $\frac{d}{dx}(\ln 3x) =$

3. $\frac{d}{dx}(\ln 2012x) =$

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Rules of Differentiation (logarithmic function)

Rule No. 6 : Chain Rule in differentiate Logarithmic function

$$\frac{d}{dx}\ln(f(x)) = \frac{d\ \ln(f(x))}{d\ f(x)} \cdot \frac{d\ f(x)}{dx} = \frac{f'(x)}{f(x)}$$

e.g. ***1.*** $\frac{d}{dx}(\ln x^2) = \left(\frac{d}{dx}x^2\right)\cdot\frac{1}{x^2} = \frac{2x}{x^2} = \frac{2}{x}$

2. $\frac{d}{dx}[\ln(2x+1)] = \left[\frac{d}{dx}(2x+1)\right]\cdot\frac{1}{2x+1} = \frac{2}{2x+1}$



3. $\frac{d}{dx}[\ln(3x+2)] =$

4. $\frac{d}{dx}[\ln(x^2-3x+2)] =$

5. $\frac{d}{dx}[\ln(x^2-1)^3] =$

6. $\frac{d}{dx}(\ln\frac{x}{3x+2}) =$

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Rules of Differentiation (logarithmic function)

Rule No. 7: **Given *a* is a positive constant and *x* > 0,**

$$\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$$

Proof:

e.g. ***1.*** $\frac{d}{dx}(\log x) = \frac{1}{x \ln 10}$



2. $\frac{d}{dx}(\log_2 x) =$

3. $\frac{d}{dx}(\log_3 3x^2) =$

4. $\frac{d}{dx}(\log_{\frac{1}{4}} \sqrt{x}) =$



5. $\frac{d}{dx}(\log_{10} \sqrt{x^2+3}) =$

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Logarithmic Differentiation（對數求導法）(I)

對於指數函數 Exponential function ($y = ka^x$)，我們並不能直接作微分，而需要首先將函數作對數 (Taking Logarithm)，然後才加以微分。這方法稱為對數求導法 (Logarithmic Differentiation)。

(I) 指數函數 ($y = ka^x$) 的對數求導法 (Logarithmic Differentiation)

e.g. Differentiate $y = a^x$ with respect to x.

Wrong approach

$$\frac{dy}{dx} = \frac{d}{dx}a^x$$
$$= xa^{x-1}$$

注意 Power Rule 是 $\frac{dy}{dx} = \frac{d}{dx}x^n = nx^{n-1}$，當指數**不是常數(Constant)**是**不能用**的!

Good approach

$$\ln y = \ln a^x$$
$$\ln y = x \ln a$$
$$\frac{d}{dx}(\ln y) = \frac{d}{dx}(x \ln a)$$
$$\frac{1}{y} \cdot \frac{dy}{dx} = \ln a \cdot \frac{d}{dx}(x) + x\frac{d}{dx}(\ln a)$$
$$\frac{1}{y} \cdot \frac{dy}{dx} = \ln a$$
$$\frac{dy}{dx} = y \ln a$$
$$\frac{dy}{dx} = a^x \ln a$$

TECHNIQUE!!

Step 1: 兩邊一齊 ***take log***

Step 2: 兩邊一齊 ***D***

最後主項變換，$\frac{dy}{dx}$做主角，變走埋個 y，搞掂！

秘訣：In 完兩邊 D 兩邊！

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Logarithmic Differentiation (I)



More examples:

1. Differentiate $y = 4\left(2^x\right)$ with respect to x.

2. Differentiate $y = x^x$ with respect to x.

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Logarithmic Differentiation（對數求導法）(2)

對數求導法除了可以用來微分指數函數外，亦可以用來微分一些冪函數 Power function ($y = kx^n$) 的，尤其是一些複雜的冪函數。

(II) 冪函數 ($y = kx^n$) 的對數求導法 (Logarithmic Differentiation)

e.g. Differentiate $y = \dfrac{x^3\sqrt{4x+7}}{x-1}$ with respect to x.

Normal approach

$$\frac{dy}{dx} = \frac{d}{dx}\frac{x^3\sqrt{4x+7}}{x-1}$$

$$= \frac{(x-1)\dfrac{d}{dx}(x^3\sqrt{4x+7}) - x^3\sqrt{4x+7}\dfrac{d}{dx}(x-1)}{(x-1)^2}$$

$$= \frac{(x-1)\left[\sqrt{4x+7}\dfrac{d}{dx}(x^3) + x^3\dfrac{d}{dx}\sqrt{4x+7}\right] - x^3\sqrt{4x+7}\dfrac{d}{dx}(x-1)}{(x-1)^2}$$

$$= \frac{(x-1)\sqrt{4x+7}(3x^2) + x^3\cdot\frac{1}{2}(4x+7)^{-\frac{1}{2}}(4) - x^3\sqrt{4x+7}}{(x-1)^2}$$

$$= .....$$

亂到唔想計！
@#$%&*$@#$~

Smart approach

$$\ln y = \ln\frac{x^3\sqrt{4x+7}}{x-1}$$

$$\ln y = \ln x^3 + \ln\sqrt{4x+7} - \ln(x-1)$$

$$\ln y = 3\ln x + \frac{1}{2}\ln(4x+7) - \ln(x-1)$$

$$\frac{1}{y}\frac{dy}{dx} = 3\cdot\frac{1}{x} + \frac{1}{2}\cdot\frac{4}{4x+7} - \frac{1}{x-1}$$

$$\frac{dy}{dx} = \frac{x^3\sqrt{4x+7}}{x-1}\left(\frac{3}{x} + \frac{2}{4x+7} - \frac{1}{x-1}\right)$$

乘除數 ln 完變加減數，
跟住仲可以變走指數，
Take log 絕對有著數！

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logarithmic Differentiation (2)



More examples:

1. If $y=\sqrt{\dfrac{(x-2)(x-5)}{(x+3)(x+6)}}$, find $\dfrac{dy}{dx}$.

2. If $y=\dfrac{e^{4x}}{(1-e^{-x})(2+e^{3x})}$, find $\dfrac{dy}{dx}$.

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HKASL Questions (Level 1)

1. ***[M&S 1996 Q.2]***

Let $y = \dfrac{\ln x}{x}$ $(x > 0)$. Find $\dfrac{dy}{dx}$ ***(2 marks)***

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HKASL Questions (Level 2)

2. ***[M&S 1995 Q.2(b)]***

If $y = \dfrac{1}{x+1}\sqrt{\dfrac{(x-2)(x+3)}{x+1}}$, use logarithmic differentiation to find $\dfrac{dy}{dx}$.

(3 marks)

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HKASL Questions (level 2)

3. ***[M&S 2001 Q.2(a)]***

Let $u = e^{2x}$ and $\frac{dy}{du} = \frac{1}{u} - 2u$. Express $\frac{du}{dx}$ and $\frac{dy}{dx}$ in terms of x.

(3 marks)

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HKASL Questions (Level 2)

4. ***[M&S 2002 Q.1]***

Let $x = -\frac{5}{t^2} + 2e^{-3t}$ and $y = \frac{10}{t^2} + e^{2t}$ $(t \neq 0)$. If $\frac{dy}{dx} = -2$, find the value of t

(6 marks)

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HKDSE Questions (level 2)

5. ***[M1 Practice Paper Q.2]***

It is given that $t = y^3 + 2y^{\frac{-1}{2}} + 1$ and $e^t = x^{x^2+1}$.

(a) Find $\frac{dt}{dy}$.

(b) By expressing t in terms of x, find $\frac{dt}{dx}$.

(b) Find $\frac{dy}{dx}$ in terms of x and y.

(5 marks)

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HKDSE Questions (level 2)

6. ***[M1 Sample Paper Q.6]***

Let $u = \sqrt{\dfrac{2x+3}{(x+1)(x+2)}}$, where $x > -1$.

(a) Use logarithmic differentiation to express $\dfrac{du}{dx}$ in terms of u and x.

(b) Suppose $u = 3^y$, express $\dfrac{dy}{dx}$ in terms of x.

(5 marks)

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HKDSE Questions (level 2)

7. ***[M1 2012 Q.4]***

Let $y = \sqrt[3]{\dfrac{3x-1}{x-2}}$, where $x > 2$.

(a) Use logarithmic differentiation to express $\dfrac{1}{y} \cdot \dfrac{dy}{dx}$ in terms of x.

(b) Using the result of (a), find $\dfrac{d^2y}{dx^2}$ when $x = 3$.

(6 marks)

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HKASL Questions (level 2)

8. ***[M&S 2005 Q.3]***

Let $w = \sqrt{\dfrac{(x-1)^3}{(x+2)(2x+1)}}$, where $x > 1$.

(a) Express $\ln w$ in the form $a\ln(x-1) + b\ln(x+2) + c\ln(2x+1)$, where a, b and c are constants. Hence find $\dfrac{dw}{dx}$.

(b) Suppose $w = 2^y$. Express $\dfrac{dy}{dw}$ in terms of w.

Hence express $\dfrac{dy}{dx}$ in terms of x. ***(7 marks)***

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